Huawei Optical Transport Failure Resolution Procedure for Access Node WHI-ACC-81

Node: WHI-ACC-81 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_7404

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Access role of the Huawei equipment at Whitefield East POP is a critical failure domain that necessitates immediate attention to prevent significant service disruptions. This failure is often precipitated by the degradation of optical signal quality, which can be attributed to various factors including but not limited to, aging of optical components, environmental factors affecting fiber integrity, or configuration mismatches. The Huawei equipment, operating with its proprietary CLI dialect, provides a comprehensive set of commands for diagnosing and resolving such failures. The ALARM_7404, categorized as a Major severity alarm, indicates a significant impact on network performance and necessitates a structured approach to diagnosis and resolution. Understanding the underlying protocol mechanics, such as the Optical Transport Network (OTN) protocol layers, and hardware specifics like the optical transceiver modules (SFP/XFP), is crucial for effective troubleshooting. This procedure outlines the necessary steps, leveraging Huawei's CLI, to identify and rectify the root cause of the Optical Transport Failure, ensuring minimal downtime and swift restoration of optical transport services.

Pre-requisite Checks

- Confirm remote hands availability at Whitefield East POP
- Verify redundant link status to ensure no single point of failure
- Check for any ongoing maintenance or recent configuration changes
- Ensure all necessary tools and spare parts are available

Potential Root Causes

Cause	Probability
SFP/XFP Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

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Command: display interface brief, Expected Output: Interface status and brief configuration, Description: This command provides a snapshot of the current interface status, helping to identify any interfaces that are down or not functioning as expected, which could indicate the source of the optical transport failure.
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Command: display optical-parameters, Expected Output: Optical parameters of the interface, Description: This command is used to display the optical parameters of the interface, such as the receive optical power and transmit optical power, which are crucial in diagnosing issues related to optical signal quality.

Command: display alarm all, Expected Output: List of all current alarms, Description: To understand the current state of the system and identify any other alarms that might be related to or contributing to the Optical Transport Failure, this command is essential.
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Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP, Command: replace sfp slot 1/0/1, Description: If the diagnostic steps indicate that the SFP/XFP is degraded, replacing it with a new one can resolve the issue. This command initiates the replacement process for the SFP/XFP in slot 1/0/1.

Step 2: Step: Repair or Replace Damaged Fiber, Command: N/A, Description: In cases where the fiber is cut or damaged, physical repair or replacement by skilled personnel is necessary. This step involves no specific CLI command but is crucial for restoring optical connectivity.

Step 3: Step: Adjust Rx/Tx Power Levels, Command: set optical-power receive -10, Description: Adjusting the receive optical power level to an appropriate range can help stabilize the optical signal. This command sets the receive optical power to -10 dBm, which may need to be adjusted based on the specific requirements of the link.

Post-Check Verification

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Command: display interface brief, Expected Output: Updated interface status, Description: After implementing the resolution steps, this command is used again to verify that the interface status has returned to normal, indicating successful resolution of the optical transport failure.

Command: display optical-parameters, Expected Output: Updated optical parameters, Description: To confirm that the optical signal quality has improved and is within acceptable ranges, this command is executed post-resolution.

Command: display alarm all, Expected Output: Updated list of alarms, Description: Finally, checking the current alarms ensures that the ALARM_7404 and any related alarms have cleared, confirming the resolution of the failure.
```

Roll-back Procedures

- Step: Revert SFP/XFP Replacement, Command: cancel replace sfp slot 1/0/1, Description: If the replacement of the SFP/XFP does not resolve the issue or introduces new problems, reverting to the original configuration is necessary. This command cancels the replacement operation.

- Step: Restore Original Fiber Configuration, Command: N/A, Description: In the event that fiber repair or replacement does not resolve the issue, restoring the original fiber configuration may be required. This involves physical actions and does not have a direct CLI command equivalent.

- Step: Reset Rx/Tx Power Levels, Command: reset optical-power, Description: If adjustments to the receive or transmit optical power levels do not improve the situation, resetting these parameters to their default values may be necessary. This command resets the optical power settings to their factory defaults.